

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VII SEMESTER

R22

VII SEMESTER		R22				
Course Code	Title of the Course	L	T	P/D	CH	C
22PC1AM401	Image Processing and Computer Vision	3	0	0	3	3
22PC1AM402	DevOps	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CS304	Cyber Security	3	0	0	3	3
22PE1AM401	Responsible AI					
22PE1IT301	Scripting Languages					
22PE1AM402	Reinforcement Learning					
22PE1AM403	Information Retrieval Systems					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Machine Learning Stream	3	0	0	3	3
	Computer Science Stream					
	Networks Stream					
	Software Engineering Stream					
	Cyber Security Stream					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2AM401	Image Processing and Computer Vision Laboratory	0	0	2	2	1
22PC2AM402	DevOps Laboratory	0	0	2	2	1
22PW4AM401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

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VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1AM404	Video Analytics	3	0	0	3	3
22PE1AM405	Cognitive Engineering					
22PE1CS404	Augmented Reality and Virtual Reality					
22PC1CY401	Blockchain Technology					
22PE1IT403	Adversarial Machine Learning					
	PROFESSIONAL ELECTIVE – VI					
22PE1AM406	Generative AI	3	0	0	3	3
22PE1DS406	Quantum Computing					
22PE1CY410	Ethical Hacking					
22PE1AM407	Robotic Process Automation					
22PE1CS405	Social Media Analysis					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4AM402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
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LR – Lab Record

B.Tech. VII Semester

(22PC1AM401) IMAGE PROCESSING AND COMPUTER VISION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Basic knowledge of Mathematics, Including Linear Algebra, Calculus and Probability; Understanding of Signals, Fourier Transforms and Basic Algorithm

COURSE OBJECTIVES:

- To understand digital image fundamentals, including sampling, quantization, and color imaging
- To apply enhancement and morphological techniques in spatial and frequency domains
- To analyze image restoration, compression, and segmentation techniques
- To explore edge, blob, line, and corner detection algorithms for vision tasks
- To implement feature detection, matching, and applications like image stitching

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the general terminology of digital image processing

CO-2: Examine the image enhancement techniques in spatial & frequency domains

CO-3: Evaluate the methodologies for image segmentation, restoration

CO-4: Apply edge, line, and corner detection techniques to extract meaningful features from images

CO-5: Implement interest point detection, feature description, and matching methods for image analysis and stitching

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	1	1	1	1	1	1	1	1	2	3	2	1
CO-2	3	3	2	2	3	1	1	1	1	1	1	2	3	1	1
CO-3	3	3	2	3	1	1	1	1	1	1	1	2	3	2	3
CO-4	2	3	3	2	3	1	1	1	1	1	1	2	2	3	3
CO-5	2	2	2	2	3	1	1	1	1	1	1	3	3	3	3

UNIT-I:

Fundamentals of Image Processing: Digital Image Fundamentals, Basic steps of Image Processing System, Sampling and Quantization of an image, the relationship between pixels, Imaging Geometry. Color Imaging.

Image Transforms: DFT, DCT, Haar.

UNIT-II:

Image Enhancement: Spatial domain methods: Histogram processing, Fundamentals of Spatial filtering, smoothing spatial filters, Sharpening spatial filters.

Frequency Domain Methods: Basics of filtering in frequency domain, image smoothing, image sharpening, Selective filtering.

Morphological Image Processing: Dilation and Erosion, Opening and Closing, the hit or miss Transformation, and some basic morphological algorithms.

UNIT-III:

Image Restoration: Image Restoration Degradation model, Algebraic approach to restoration, Inverse Filtering, Least Mean square filters.

Image Compression: Image compression fundamentals - Coding Redundancy, Spatial and Temporal redundancy, Compression models: Lossy and Lossless, Huffman coding, Arithmetic coding, LZW coding, Run length coding, Bit plane coding, Transform coding, Predictive coding, JPEG Standards.

Image Segmentation: Segmentation concepts, Thresholding, Region Based segmentation.

UNIT-IV:

Introduction to Computer Vision and its Applications: A brief history, Edge Detection: Difference of Gaussian (DoG), Gradient magnitude, Canny Edge Detector; Blob Detector: Laplacian of Gaussian; Lines Detection: Hough transform algorithm; Corner Detection: Harris Corner Detector.

UNIT-V:

Interest Point Detection: Introduction

Feature Descriptors and matching: Histogram of Oriented Gradients (HoG), Local Binary Patterns (LBP), Scale Invariant Feature Transform (SIFT); Feature Matching: RANSAC algorithm, Application: Image Stitching; Visual Matching: Bag-of-words.

TEXT BOOKS:

1. Digital Image Processing, Rafael C. Gonzalez and Richard E. Woods, 3rd Edition, Pearson, 2008
2. Computer Vision: Algorithms and Applications, Richard Szeliski Springer, 2010 (<http://szeliski.org/Book/>)

REFERENCES:

1. Digital Image Processing, S. Sridhar, 2nd Edition, Oxford, 2016
2. Digital Image Processing, William K. Pratt, 3rd Edition, John Willey, 2004
3. Computer Vision: A Modern Approach, Forsyth and Ponce, Prentice Hall, 2002
4. Programming Computer Vision using Python (http://programmingcomputervision.com/downloads/ProgrammingComputerVision_CCdraft.pdf)

B.Tech. VII Semester

(22PC1AM402) DEVOPS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce Agile methodologies and DevOps principles, highlighting their role in improving software development efficiency and addressing bottlenecks
- To explore modern software architectures, emphasizing the transition from monolithic to microservices and serverless models while integrating DevOps practices
- To develop understanding in version control and containerization, utilizing Git, GitHub, Docker, and Kubernetes for efficient project collaboration and deployment
- To familiarize with CI/CD and automation skills, covering Jenkins, CircleCI, GitHub Actions, infrastructure as code (Terraform), testing and monitoring tools
- To introduce DevSecOps and security best practices, focusing on secure coding, secret management, container security, and automated security testing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse and apply DevOps principles to enhance business agility, and address software development bottlenecks

CO-2: Design and manage software architectures, including monolithic, microservices and serverless using DevOps practices

CO-3: Implement version control and collaboration using Git and GitHub, and manage containerized applications using Docker and Kubernetes

CO-4: Build and manage CI/CD pipelines using Jenkins, CircleCI, and Github Actions for automated deployments and testing

CO-5: Understand IaC and the need to integrate security in DevOps with DevSecOps and GitOps to ensure secure and efficient software delivery

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	2	2	2	3	3	2	3
CO-2	3	3	3	2	3	2	2	1	1	2	2	3	3	2	3
CO-3	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-4	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

UNIT-I:

Introduction to DevOps & Business Agility: Introduction to DevOps (What is DevOps, benefits and goals of DevOps, DevOps lifecycle and phases), DevOps and Business Agility (What is business agility, DevOps's role in enhancing agility, collaboration between dev, QA and ops), Bottlenecks (Identifying bottlenecks in software delivery, common bottlenecks, how DevOps addresses bottlenecks).

UNIT-II:

Software Architectures & the Need for DevOps: Introduction to Agile (Agile vs Waterfall model, Agile methodologies – Scrum, Kanban, Continuous Integration and Continuous Development – CI/CD overview, how Agile and DevOps work together), Software Architectures (Importance of separation of concerns, different types of architectures and examples, Monolithic vs Microservices vs Serverless, benefits and challenges of each)

UNIT-III:

Project Management with Version Control, Containerization and Kubernetes: Version Control Systems (Introduction to VCS, Types of VCS, Git vs other VCS tools, Important Git commands, GitHub for collaboration – branching and merging, pull requests, code reviews), Containerization with Docker (What are containers, Docker vs Virtual Machines, building and running containers, port mapping, container networking overview, DockerHub introduction), Kubernetes (What is Kubernetes, core concepts- pods, nodes, deployments, services, Configmaps and secrets, Kubernetes networking overview, scaling applications (horizontal, vertical), persistent storage, deploying applications- kubectl, config files)

UNIT-IV:

CI/CD, Testing and Monitoring: Importance of CI/CD, Jenkins overview (pipelines, build dependencies, plugins, triggers), CI/CD with CircleCI, Using Github Actions for CI/CD, Testing and Automation (importance of automated testing, types of testing – unit, integration, end-to-end, Test Driven Development (TDD), API testing using Postman, JUnits/PyTests), Monitoring and Logging (Importance of logging and monitoring in DevOps, monitoring tools – Prometheus, Grafana, logging tools – Splunk, New Relic, implementing alerts and logging analysis)

UNIT-V:

Infrastructure as Code(IaC), GitOps and DevSecOps Overview: What is IaC and Terraform, writing Terraform configurations, Terraform State Management and Kubernetes infrastructure, GitOps and ArgoCD (brief introduction to GitOps, ArgoCD setup and declarative deployments), DevSecOps – Security in DevOps (Shift-left security & container image scanning, secret management – vault, AWS/Azure secret manager, Automating security in CI/CD)

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, 2018
2. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint, Wiley

REFERENCE:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley

B.Tech. VII Semester

(22PE1CS304) CYBER SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To summarize various types of cyber-attacks and cyber-crimes
- To understand cyber laws and the concepts of digital forensics
- To discuss safety measures for the protection of mobile and wireless devices
- To learn the organizational security implications and threats
- To study the impact of data privacy attacks on various domains

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the need of cyber security and various types of attacks

CO-2: Understand national and international regulations of cyber security and cyber forensics

CO-3: Interpret the security challenges related to mobile and wireless devices

CO-4: Analyze the security and privacy implications of an organization

CO-5: Examine the data privacy concepts and cybercrime in different domains

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	2	2	2	-	3	2	2	-	2	2	3	3
CO-2	1	1	3	3	1	1	-	-	3	3	-	2	1	3	3
CO-3	2	2	3	2	2	2	-	-	3	3	-	3	2	3	2
CO-4	2	2	3	2	3	2	-	3	3	3	2	3	2	2	3
CO-5	1	2	2	3	3	3	-	3	3	3	2	3	3	3	2

UNIT-I:

Introduction to Cyber Security: Basic Cyber Security Concepts, layers of security, Vulnerability, threat, Harmful acts, Internet Governance – Challenges and Constraints, Computer Criminals, CIA Triad, Assets and Threat, motive of attackers, active attacks, passive attacks, Software attacks, hardware attacks, Cyber Threats-Cyber Warfare, Cyber Crime, Cyber terrorism, Cyber Espionage, etc., Comprehensive Cyber Security Policy.

UNIT-II:

Cyberspace and the Law & Cyber Forensics: Introduction, Cyber Security Regulations, Roles of International Law. The INDIAN Cyberspace, National Cyber Security Policy, Historical background of Cyber forensics, Digital Forensics Science, The Need for Computer Forensics, Cyber Forensics and Digital evidence, Forensics Analysis of Email, Digital Forensics Lifecycle, Forensics Investigation, Challenges in Computer Forensics

UNIT-III:

Cybercrime: Mobile and Wireless Devices: Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless

Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones, Organizational security Policies and Measures in Mobile Computing Era, Laptops.

UNIT-IV:

Cyber Security: Organizational Implications: Introduction, cost of cybercrimes and IPR issues, web threats for organizations, OWASP top 10 web application security risks, social media marketing: security risks and perils for organizations, social computing and the associated challenges for organizations

UNIT-V:

Privacy Issues: Basic Data Privacy Concepts: Fundamental Concepts, Data Privacy Attacks, Data linking and profiling, privacy policies and their specifications, privacy policy languages, privacy in different domains- medical, financial, etc Cybercrime: Examples and Mini-Cases Examples: Official Website of Maharashtra Government Hacked, Indian Banks Lose Millions of Rupees, Parliament Attack, Pune City Police Bust Nigerian Racket, e-mail spoofing instances. Mini Cases: The Indian Case of Online Gambling, An Indian Case of Intellectual Property Crime, Financial Frauds in Cyber Domain.

TEXT BOOKS:

1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Nina Godbole and Sunit Belpure, Wiley
2. Computer and Cyber Security: Principles, Algorithm, Applications, and Perspectives, B. B. Gupta, D. P. Agrawal, Haoxiang Wang, CRC Press

REFERENCES:

1. Mining of Massive Datasets, v2.1, Jure Leskovek, Anand Rajaraman, Jeffrey Ullman, Cambridge University Press, 2014
2. Data Science from Scratch: First Principles with Python, Joel Grus, 1st Edition, 2015
3. Doing Data Science, Straight Talk from the Frontline, Cathy O'Neil, Rachel Schutt, O' Reilly, 1st Edition, O'Reilly, 2013
4. Mining of Massive Datasets, Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman, 2nd Edition, Cambridge University Press, 2014

B.Tech. VII Semester

(22PE1AM401) RESPONSIBLE AI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the fundamental principles of responsible AI and the associated metrics
- To understand the potential risks and advantages linked with responsible AI
- To illustrate instances of dimensions related to Responsible AI and their practical implementations
- To provide effective communication of ethical considerations of AI and ML effectively to various stakeholders
- To understand comprehensive understanding of ethical challenges in AI and ML

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse ethical AI principles and techniques in modeling diverse AI and ML models

CO-2: Establish connections between different responsible AI frameworks and the principles of governance

CO-3: Analyse the principles of enterprise risk handling techniques for projects involving AI and ML

CO-4: Apply the concepts of fairness assessment and bias mitigation for real-time applications

CO-5: Evaluate ethical theories and frameworks applicable to AI and ML-related applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	1	1	3	2	3	1	1	1	1	2	1	2
CO-2	1	1	1	1	2	3	2	3	1	1	1	2	1	1	1
CO-3	2	1	1	1	1	3	1	3	1	1	1	2	2	2	3
CO-4	1	1	1	1	1	2	2	3	1	1	1	1	1	1	2
CO-5	2	1	1	1	1	3	2	3	1	1	1	1	2	1	3

UNIT- I:

Explainable and Ethical AI Primer

The imperative of AI governance, Key terminologies: Explainability, Interpretability, Explicability, Safe and trustworthy, Fairness, Ethics. Enterprise risk management and governance: Tools for enterprise risk governance, AI risk governance in the enterprise.

UNIT-II:

AI in hiring and Recruitment: Facial recognition, Bias in large language models (LLMs), AI-powered inequity and discrimination, Policing and surveillance, social media and

attention engineering, The environmental impact, Autonomous weapon systems and military.

UNIT-III:

Enterprise Risk Observability Model Governance: Opening The Algorithmic Black Box, Getting started with interpretable methods. The business case for explainable AI, Taxonomy of ML explainability methods.

Enterprise Starter Kit for Fairness, Accountability, And Transparency. Getting started with enterprise AI governance. The indispensable role of the C-suite in fostering responsible AI adoption. The role of internal AI boards in enterprise AI governance. Ethical AI upskilling and education.

UNIT-IV:

Fairness in AI Systems with Microsoft Fair learn: Getting started with fairness, Fairness metrics, Fairness-related harms, Getting started with Fair learn. Fairness Assessment and Bias Mitigation with Fair learn and the Responsible AI Toolbox: Fairness metrics, Bias and disparity mitigation with Fair learn, The Responsible AI Toolbox.

UNIT-V:

Foundational Models and Azure OpenAI: Foundation models, Enterprise use of foundation models and bias remediation, Azure OpenAI, Data, privacy, and security for Azure OpenAI Service, Azure OpenAI Service models.

Interpretability Toolkits And Fairness Measures – AWS, GCP, Azure, And AIF 360. Getting started with hyperscaler interpretability toolkits, Google Vertex Explainable AI. AWS Sagemaker Clarify, Azure Machine Learning model interpretability. Open source toolkits and lenses.

TEXT BOOKS:

1. Responsible AI in the Enterprise, Adnan Masood, Heather Dawe, Dr. Ehsan Adeli, Packt Publishing, 2023
2. Machine Learning for High-Risk Applications, Patrick Hall, James Curtis, Parul Pandey, O'Reilly Media, 2023

REFERENCES:

1. Responsible AI Architect's Guide Prescribing Responsible AI Best Practices Implementation Methods and Tools for Enterprise Adoption (architect-guide.pdf (indiaai.gov.in))
2. Responsible AI: Best Practices for Creating Trustworthy AI Systems By CSIRO, Qinghua Lu, Jon Whittle, Xiwei Xu, Liming Zhu
3. Responsible AI: Implementing Ethical and Unbiased Algorithms, Sray Agarwal, Shashin Mishra, 1st Edition, 2021
4. Responsible AI - my.UQ - The University of Queensland, Australia

B.Tech. VII Semester

(22PE1IT301) SCRIPTING LANGUAGES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Knowledge of Programming Languages

COURSE OBJECTIVES:

- To learn the concepts scripting languages
- To learn the various features of PERL
- To learn about PHP basics
- To learn PHP features and advanced concepts
- To learn how to use MySQL with PHP

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the role of scripting languages and PERL in the real world

CO-2: Apply the concepts of PERL

CO-3: Understand the use of PHP in the field of application development

CO-4: Apply the advanced features of PHP in building applications

CO-5: Design and Develop the applications using backend databases like MySQL

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)			
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3	PSO-4
CO-1	3	3	3	3	2	2	-	2	2	2	-	-	3	-	2	-
CO-2	3	-	3	3	3	3	2	-	-	-	-	-	3	2	2	2
CO-3	-	2	3	-	-	3	-	-	-	-	2	3	3	3	2	3
CO-4	3	3	-	3	-	-	2	-	-	-	3	3	3	2	-	-
CO-5	3	3	3	3	3	3	-	2	2	-	3	3	3	3	2	3

UNIT-I:

Introduction to PERL and Scripting: Scripts and Programs, Origin of Scripting, Scripting Today, Characteristics of Scripting Languages, Uses for Scripting Languages, Web Scripting, The universe of Scripting Languages.

PERL: Introduction, Scalar Data, Lists and Arrays, Subroutines, Input and Output Hashes

UNIT-II:

Advanced PERL: Finer points of looping, pack and unpack, file system, eval, data structures, packages, modules, objects, subroutines. Pattern and regular expressions, processing text with regular expressions

UNIT-III:

PHP Basics: PHP Basics Features, Embedding PHP Code in your Web pages, Outputting the data to the browser. Data types, Variables, Constants, Expressions, String

interpolation, Control Structures. Function, Creating a Function, Function Libraries, Arrays, Strings and Regular Expressions.

UNIT-IV:

Advanced PHP: Advanced PHP Programming Php and Web Forms, Files, PHP Authentication and Methodologies-Hard Coded, File Based, Database Based, IP Based, Login Administration, Uploading Files with PHP, Sending Email using PHP, PHP Encryption Functions, the Mcrypt package, Building Web sites for the World.

UNIT-V:

PHP and Database: PHP and MySQL: Basic commands with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names, creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, PHP myadmin and database bugs.

TEXT BOOKS:

1. The World of Scripting Languages, David Barron, Wiley
2. Perl by Example, E. Quigley, Pearson Education

REFERENCES:

1. Programming Perl, Larry Wall, T. Christiansen and J. Orwant
2. Beginning PHP and MySQL, Jason Gilmore, 3rd Edition, Apress Dreamtech

B.Tech. VII Semester

(22PE1AM402) REINFORCEMENT LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Machine Learning

COURSE OBJECTIVES:

- To understand multi-armed bandits
- To introduce finite Markov decision processes
- To understand dynamic programming and evaluation
- To understand Monte Carlo methods and decision process
- To analyze temporal difference learning and policy control

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Summarize the elements of Reinforcement Learning and apply K-Armed bandit problem for different situations

CO-2: Apply finite Markov decision process and model a control task in the Markov decision process framework

CO-3: Apply dynamic programming for policy evaluation and improvement

CO-4: Outline the Monte Carlo methods and sampling techniques

CO-5: Apply temporal difference for prediction models

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	2	-	-	-	-	-	2	2	2	2
CO-2	1	2	2	1	-	-	-	-	-	-	-	2	1	2	2
CO-3	2	2	3	-	2	1	-	-	-	-	-	2	2	1	2
CO-4	2	1	3	1	-	-	-	-	-	-	-	2	3	3	3
CO-5	2	2	3	-	1	1	-	-	-	-	-	2	3	3	3

UNIT-I:

Introduction to Reinforcement Learning: Reinforcement Learning, Elements of Reinforcement Learning, Limitations and Scope.

Multi-Armed Bandits: A K-Armed Bandit Problem, Action-Value Methods, 10-Armed Testbed, Incremental Implementation, Tracking a Nonstationary Problem, Optimistic Initial Values, Upper Confidence Bound Action Selection, Gradient Bandit Algorithm, Associative Search.

UNIT-II:

Finite Markov Decision Processes: Agent-Environment Interface, Goals and Rewards, Returns and Episodes, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions, Optimal Policies and Optimal Value Functions, Optimality and Approximation.

UNIT-III:

Dynamic Programming: Policy Evaluation, Policy Improvement, Policy Iteration, Value Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming.

UNIT-IV:

Monte Carlo Methods: Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control, Monte Carlo Control Without Exploring Starts, Off-Policy Prediction via Importance Sampling, Incremental Implementation, Off-Policy Monte Carlo Control, Discounting Aware Importance Sampling, Per- Decision Importance Sampling.

UNIT-V:

Temporal Difference Learning: Temporal Difference (TD) Prediction, Advantages of TD Prediction Models, Optimality of TD(0), Sarsa On-Policy TD Control, Q-Learning Off-Policy TD Control, Expected Sarsa, Maximization Bias and Double Learning, Games, Afterstates and Other Special Cases.

TEXT BOOK:

1. Reinforcement Learning: An Introduction, Richard S. Sutton, Andrew G. Barto, 2nd Edition, MIT Press

REFERENCE:

1. Reinforcement Learning, Phil Winder, 1st Edition, O'Reilly

ONLINE RESOURCES:

1. <https://www.microsoft.com/en-us/research/event/reinforcement-learning-day-2019/agenda/>
2. <https://deepmind.google/>
3. <https://web.stanford.edu/class/cs234/index.html>
4. <https://web.mit.edu/dimitrib/www/RLbook.html>
5. <https://www.codementor.io/reinforcement-learning-experts>

B.Tech. VII Semester

(22PE1AM403) INFORMATION RETRIEVAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To describe the differences between repositories like Database Management Systems, Information retrieval systems and data warehouse
- To discover various pre-processing techniques which can apply on text documents to outline the structure of queries and documents
- To articulate fundamental functions used in information retrieval such as automatic indexing, abstracting, clustering and visualizing data
- To learn the important concepts, algorithms, and data/file structures that are necessary to specify, design, and implement Information Retrieval (IR) systems
- To explore various text search algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and understand the relationships between various Repository Systems

CO-2: Apply knowledge of data structures and indexing methods in information retrieval Systems

CO-3: Implement various clustering, searching techniques and information visualization techniques

CO-4: Analyze clustering techniques and algorithms using evaluation measures

CO-5: Implement various text search algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	2	3	3	2	1	1	2	2	2	3	3	3
CO-2	3	3	3	3	3	2	2	2	2	2	2	2	3	3	3
CO-3	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-4	3	3	3	3	3	3	2	2	2	3	3	2	3	3	3
CO-5	2	2	2	2	3	3	2	1	1	2	2	2	3	3	3

UNIT-I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses, Information Retrieval System Capabilities – Search, Browse, Miscellaneous.

UNIT-II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N – gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT-III:

Automatic Indexing: Classes of automatic indexing, Statistical Indexing, Natural language, Concept indexing, Hypertext linkages.

Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT-IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean Systems, Searching the Internet and hypertext.

UNIT-V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems.

Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXT BOOKS:

1. Information Storage and Retrieval Systems: Theory and Implementation, Kowalski, Gerald, Mark T. Maybury, Springer
2. Modern Information Retrieval: Ricardo Baeza, Yates, Pearson Education, 2007

REFERENCES:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman and Ophir Frieder, 2nd Edition, Springer
2. Information Retrieval Data Structures and Algorithms, Frakes W. B., Ricardo Baeza-Yates, Prentice Hall, 1992

B.Tech. VII Semester

(22PC2AM401) IMAGE PROCESSING AND COMPUTER VISION LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Proficiency in Python programming and familiarity with libraries like NumPy and OpenCV

COURSE OBJECTIVES:

- To understand and implement matrix operations and image attributes using Python
- To apply transformation, enhancement, and filtering techniques in image processing
- To analyze and implement segmentation, edge detection, and morphological operations
- To extract features using HOG, SIFT, and LBP for detection and recognition tasks
- To develop image processing applications like compression, stitching, and classification

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Implement fundamental operations using Python libraries like SciPy for matrix arithmetic and image attribute extraction

CO-2: Apply 2D and 3D transformations, image enhancement techniques, and filtering methods for image processing tasks

CO-3: Analyze advanced algorithms for image segmentation, edge detection, and morphological operations

CO-4: Extract meaningful features using techniques like HOG, SIFT, and LBP to solve real-world problems like object detection and face recognition

CO-5: Develop applications for image compression, stitching, and classification using advanced methods such as Bag of Words and panorama creation

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	3	1	1	1	1	1	1	2	3	2	1
CO-2	3	3	3	2	3	1	1	1	1	1	1	2	3	2	2
CO-3	2	3	3	3	3	1	1	1	1	1	1	2	3	3	3
CO-4	2	3	3	2	3	1	1	1	1	1	1	2	3	3	3
CO-5	2	2	3	2	3	1	1	1	1	1	1	3	3	3	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1: Introduction to SCIPY and Python code to perform arithmetic operations on matrices

WEEK 2: Program to extract different attributes of an image and image negation

WEEK 3: Write a program to perform different 2D and 3D transformation techniques on a given image using Python

WEEK 4: Program to implement image enhancement using various filters (smoothing and sharpening) and histogram

WEEK 5: Write a program to simulate different Linear and Non-Linear filters on a given image using Python

WEEK 6: Program to implement image segmentation algorithm using Python

WEEK 7: Program for Image Compression

WEEK 8: Program for morphological operation opening and closing.

WEEK 9: Write a program to implement Canny edge detection algorithm using Python

WEEK 10: Write a program to implement HOG descriptor for person detection using Python

WEEK 11: Write a program to extract interest points of an image by SIFT Detector using Python

WEEK 12: Write a program to implement Image stitching panorama using Python

WEEK 13: Write a program to implement LBP for face detection a given image using Python.

WEEK 14: Write a program to perform Image Classification using Bag of Word

TEXT BOOKS:

1. Digital Image Processing, Rafael C. Gonzalez and Richard E. Woods, 3rd Edition, Pearson, 2008
2. Computer Vision: Algorithms and Applications, Richard Szelisky Springer, 2010 (<http://szeliski.org/Book/>)

REFERENCES:

1. Digital Image Processing, S. Sridhar, 2nd Edition, Oxford, 2016
2. Digital Image Processing, William K. Pratt, 3rd Edition, John Willey, 2004
3. Computer Vision: A Modern Approach, Forsyth and Ponce, Prentice Hall, 2002
4. Programming Computer Vision using Python (http://programmingcomputervision.com/downloads/ProgrammingComputerVision_CCdraft.pdf)
5. Feature Extraction & Image Processing for Computer Vision, Mark and Alberto S. Aquando, 3rd Edition, Academic Press, 2012

B.Tech. VII Semester

(22PC2AM402) DEVOPS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Proficiency in at least one programming language (e.g., Python, Java, or Shell scripting), Basic understanding of REST APIs and software development concepts

COURSE OBJECTIVES:

- To provide hands-on experience in developing backend applications, implementing CRUD operations, and exposing REST APIs
- To introduce version control systems using Git and GitHub, enabling effective source code management and collaboration
- To familiarize with continuous integration and deployment (CI/CD) pipelines, using tools like CircleCI and GitHub Actions
- To equip with containerization skills, enabling them to package and manage applications using Docker and Kubernetes
- To develop proficiency in API testing and automation, utilizing tools like Postman and writing test cases for backend services

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Develop backend applications with CRUD operations and deploy them following DevOps best practices

CO-2: Manage and collaborate on source code using Git and GitHub, demonstrating version control strategies

CO-3: Set up and utilize CI/CD pipelines using CircleCI/GitHub Actions to automate builds and deployments

CO-4: Containerize applications using Docker, automate deployments, and orchestrate them using Kubernetes

CO-5: Perform API testing with Postman and write automated test cases to validate backend functionality

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	1	1	2	2	2	3	3	2	3
CO-2	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-3	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-4	3	3	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. Write code for a simple backend project. It should have basic CRUD operations and APIs (GET, POST, DELETE, PATCH/UPDATE).
2. Explore Git and GitHub commands.
3. Practice Source code management on GitHub. Experiment with the source code written in exercise 1.
4. Explore Docker commands for container management.
5. Develop a simple containerized application using Docker.
6. Integrate Kubernetes and Docker
7. Automate the process of running containerized application developed in exercise 7 using Kubernetes.
8. CircleCI/ Github Actions setup, explore the environment.
9. Demonstrate continuous integration and development using CircleCi/Github Actions
10. Install postman and explore manual API testing.
11. Develop test cases for simple APIs in the language of choice.
12. Integrate any of the suitable monitoring and logging dashboards with your application.
13. Hands-on parallel exploration of Terraform, and its configurations

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, 2018
2. DevOps Tools from Practitioner's Viewpoint, Deepak Gaikwad, Viral Thakkar, Wiley

REFERENCES:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley
2. Edureka DevOps Full Course - https://youtu.be/S_0q75eD8Yc

B.Tech. VIII Semester

(22PE1AM404) VIDEO ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To learn the fundamentals of digital image processing, as well as image and video analysis
- To learn techniques for extracting information from videos
- To analyze video processing techniques and their applications in video analytics and surveillance systems
- To understand and apply models used for human action recognition in images and video
- To identify and track humans in videos and apply gait analysis techniques for recognition

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Design basic digital signal processing systems

CO-2: Construct motion estimation algorithms for video analysis

CO-3: Develop techniques for video detection and tracking

CO-4: Analyze and recognize activity patterns and behaviors in video data

CO-5: Apply face recognition algorithms to identify humans in images and videos

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	3	-	-	-	2	-	-	-	2	-	3	-
CO-2	3	2	3	3	3	2	2	3	2	1	3	3	3	-	3
CO-3	2	2	2	3	-	-	-	2				2		3	-
CO-4	3	3	2	3	1	1	1	2	1	1	1	3	2	3	3
CO-5	3	3	3	3	-	-	-	1	1	1	2	2		3	-

UNIT-I:

Introduction: Multidimensional signals and systems: Signals, Transforms, Systems, Sampling theorem, Human visual system and color, Digital video, 3D video, Digital-video applications, Image and video quality.

UNIT-II:

Motion Estimation: Image formation, Motion models, 2D Apparent Motion estimation, Differential methods, Matching methods, Non-linear optimization methods, Transform-domain methods, 3D motion and structure estimation.

UNIT-III:

Motion Tracking: Image Segmentation, Change Detection, Motion Segmentation, Motion Tracking: Graph-based tracking, Lucas-Kanade tracker, Particle filtering, Image and Video Matting, Performance Evaluation

UNIT-V:

Behavioral Analysis and Activity Recognition: Event Modelling, Behavioural Analysis, Human Activity Recognition, Complex Activity Recognition, Activity modelling using 3D shape, Video summarization, Shape-based activity models, Suspicious Activity Detection.

UNIT-V:

Human Face Recognition & Gait Analysis: Overview of Human Recognition algorithms, **Human Recognition using Face:** Face Recognition from still images, Face Recognition from video, Evaluation of Face Recognition Technologies, Human Recognition using gait: HMM Framework for Gait Recognition, View Invariant Gait Recognition, Role of Shape and Dynamics in Gait Recognition.

TEXT BOOKS:

1. Digital Video Processing, Murat Tekalp, 2nd Edition, Pearson, 2015
2. Recognition of Humans and Their Activities Using Video, Rama Chellappa, Amit K. Roy-Chowdhury, S. Kevin Zhou

REFERENCES:

1. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2011
2. Video Processing and Communications, Yao Wang, Jorn Ostermann and Ya-Qin Zhang, Prentice Hall, 2001
3. Background Modeling and Foreground Detection for Video Surveillance: Traditional and Recent Approaches, Implementations, Benchmarking and Evaluation, Thierry Bouwmans, Fatih Porikli, Benjamin Höferlin and Antoine Vacavant, CRC Press, Taylor and Francis, 2014
4. Computer Vision and Action Recognition-A Guide for Image Processing and Computer Vision Community for Action Understanding, Md. Atiqur Rahman Ahad, Atlantis Press, 2011

B.Tech. VIII Semester

(22PE1AM405) COGNITIVE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify the basics of cognitive science engineering with focus on mind organization, brains
- To analyze human intelligence and artificial intelligence
- To understand basics of neuroscience and linguistics
- To analyze various levels of cognitive functioning and its applications
- To relate the concepts of cognitive field in robotics and real time applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of computers in cognitive science and neuroscience applications

CO-2: Illustrate various information processing models and cognitive architecture

CO-3: Apply various computation tools for cognitive functioning

CO-4: Analyze cognitive science applications in image processing linguistic applications

CO-5: Demonstrate the use of cognitive models and patterns in distributed environment, and real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	2	1	1	1	2	1	1	1	2	2	2
CO-2	2	3	2	2	3	1	1	2	2	2	1	1	3	3	2
CO-3	2	3	3	2	2	1	2	1	2	2	2	1	3	3	3
CO-4	3	3	3	2	2	2	1	2	3	3	2	2	3	3	2
CO-5	2	2	2	3	2	1	2	1	2	2	3	2	2	2	3

UNIT-I:

Introduction to Cognitive Science: The Cognitive view –Some Fundamental Concepts – Computers in Cognitive Science – Applied Cognitive Science – The Interdisciplinary Nature of Cognitive Science, The Architecture of the Mind. Cognitive Processes, Computational Neuroscience - The Organization of the mind.

UNIT-II:

Higher-Level Cognition and Challenges: Dynamical systems and situated cognition- Challenges – Emotions and Consciousness – Computation of Cognitive Functioning in machines: Robotics, Human-Robotics Interaction.

UNIT-III:

Cognitive Science for Vision/Image Processing: Perception and sensing: visual cognition, cognitive mechanisms of vision, Feature Extraction from Images,

Information Processing in Perception and Visual Behavior, Cognitive function measurement tools and software.

UNIT-IV:

Language Acquisition and Semantics: Milestones in acquisition, Theoretical perspectives, Semantics and Cognitive Science, Meaning and entailment, Sense, Problems in possible worlds Semantics, Cognitive and computational models of semantic Processing.

UNIT-V:

Distributed Cognitive Learning Using Fuzzy Cognitive Maps: Axelrod's cognitive maps, Kosko's model, adaptive FCM, Pal and Konar's FCM model-examples.

TEXT BOOKS:

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2nd Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind José Luis Bermúdez, Cambridge University Press, 2010
3. Cognitive Engineering: A Distributed Approach to Machine Intelligence (Advanced Information and Knowledge Processing), Amit Konar, Springer, 2005

REFERENCES:

1. Computational Vision: Information Processing in Perception and Visual Behavior, Hanspeter A. Mallot, Translated by John S. Allen
2. How the Mind Works, Steven Pinker, 2009
3. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
4. Mind: Introduction to Cognitive Science, Paul Thagard, 2nd Edition, MIT Press, 2004
Wilson Robert A., & Keil Frank C. (Eds.)
5. The MIT Encyclopedia of the Cognitive Sciences (MITECS), MIT Press, 2001

B.Tech. VIII Semester

(22PE1CS404) AUGMENTED REALITY AND VIRTUAL REALITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming and Data Structures

COURSE OBJECTIVES:

- To understand the basic concepts of Virtual and Augmented Reality
- To identify the differences in AR/VR concepts and technologies
- To understand the fundamental concepts relating to Virtual Reality such as presence, immersion, and engagement
- To evaluate usability of AR/VR applications and critique their use of AR/VR capabilities
- To design and prototype effective AR/VR applications using UNITY platform for business, industry, non-profit and government organizations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Introduce role of augmented and virtual reality in interactive application development

CO-2: Apply Geometric Modeling Techniques

CO-3: Implement virtual environment with respect to user experience

CO-4: Design Virtual/Augmented Reality Applications

CO-5: Acquire fundamental and hands-on skills needed to develop AR/VR applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	1	1	2	2	1	-	2	2	3	3	3
CO-2	1	1	1	3	1	2	2	1	2	1	1	2	3	3	3
CO-3	1	1	2	-	3	1	1	1	1	2	1	2	3	3	3
CO-4	-	2	-	-	-	2	-	2	1	1	3	1	3	3	3
CO-5	-	-	2	-	3	-	-	-	-	-	-	-	3	3	3

UNIT-I:

Introduction to Augmented Reality (AR): Definition and Scope A Brief History of Augmented Reality Displays (Multimodal Displays, Spatial Display Model, and Visual Displays) Strong vs Weak AR Applications of AR Challenges in AR

UNIT-II:

Introduction to Virtual Reality (VR): Definition and Scope Types of VR Characteristics of VR Basic VR environments Limitations of VR environments Immersion Vs Presence Key hardware requirements for VR

UNIT-III:

Interaction Design for AR/VR Environments: Interaction design process Identifying user needs AR/VR design considerations Typical AR/VR Interface Metaphors Affordances in AR/VR Human Information Processing Design for Perception and cognition User experience (UX) guidelines for AR/VR UX challenges for AR/VR Prototyping for AR/VR Evaluation of the developed AR/VR prototype

UNIT-IV:

Unity Overview: Windows, Interface, Navigation, Terminology, Game Objects, Hierarchy, Parenting Objects Asset Store, Importing Plug-ins Creating a Terrain, Materials, Colors, Transparency Introduction to MonoBehaviour: Awake, Start, Update

UNIT-V:

Introduction to Vuforia and Physics in UNITY

Vuforia Overview: Interface, Navigation, Terminology, Image Targeting, Custom Images Overview of Physics in Unity

Introduction to Scripting: Terminology, Creating Objects, Accessing Components, Debugging, Lists, Loops

TEXT BOOKS:

1. Understanding virtual reality, Sherman W. R., & Craig A. B., Morgan Kauffman, 2003
2. Augmented Reality: Principles and Practice, Schmalstieg D., & Hollerer T., Addison-Wesley Professional, 2016

REFERENCES:

1. A survey of augmented reality. Presence: Teleoperators & Virtual Environments, Azuma R. T., 1997, 6(4), 355–385
2. Recent Advances in Augmented Reality, IEEE Computer Graphics and Applications, Azuma R., Baillot Y., Behringer R., Feiner S., Julier S., & MacIntyre B., 2001, 21(6), 34-47
3. A Cost-Effective Interactive 3D Virtual Reality System Applied To Military Live Firing Training, Virtual Reality, Bhagat K. K., Liou, W. K., & Chang, C. Y, 2016, 20(2), 127-140. doi:10.1007/s10055-016-0284-x
4. Augmented Reality Technologies, Systems And Applications, Multimedia Tools and Applications, Carmigniani J., Furht B., Anisetti M., Ceravolo P., Damiani E., & Ivkovic M., 2011, 51(1), 341-377
5. Human Augmentation: Past, Present and Future. International Journal Of Human Computer Studies, Raisamo R., Rakkolainen I., Majaranta P., Salminen K., Rantala J., & Farooq, 2019, 131, 131-143. doi: <https://doi.org/10.1016/j.ijhcs.2019.05.008>

ONLINE RESOURCES:

1. Virtual Reality Engineering, Manivannan M., 2018, IIT Madras, <https://nptel.ac.in/courses/121106013>
2. Industry 4.0: Augmented Reality and Virtual Reality, Misra S., 2019, IIT Kharagpur, <https://www.youtube.com/watch?v=zLMgdYI82IE>
3. Augmented Reality - Fundamentals and Development, NPTEL Special Lecture Series, Dube A., 2020, <https://www.youtube.com/watch?v=MGuSTAqIZ9Q>
4. <http://cambum.net/course-2.htm>

B.Tech. VIII Semester

(22PC1CY401) BLOCKCHAIN TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an understanding of blockchain technology, including its origin, components, types, consensus mechanisms, and cryptocurrency applications
- To explore public blockchain systems and smart contracts, their principles, implementations, and industry applications
- To know the security issues of blockchain technology
- To understand security challenges in blockchain and explore industry applications.
- To study practical blockchain implementations through case studies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze blockchain technology by examining its origin, components, types, consensus mechanisms, and cryptocurrency applications

CO-2: Evaluate public blockchains and smart contracts, their principles, implementations, and industry applications

CO-3: Examine private and consortium blockchains, their applications, and the role of ICOs in blockchain fundraising

CO-4: Apply blockchain security principles and explore its applications, limitations, and future scope

CO-5: Implement blockchain solutions through case studies and development with Python and Hyperledger Fabric

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	1	1	1	2	1	1	1	2	3	1	1
CO-2	3	2	3	3	3	2	2	3	2	1	1	3	3	1	3
CO-3	2	2	2	1	1	1	1	2	1	3	1	2	1	3	1
CO-4	3	3	2	1	1	1	1	2	1	3	1	3	1	3	3
CO-5	3	3	3	1	1	1	1	1	1	3	1	2	1	3	1

UNIT-I:

Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency – Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.

UNIT-II:

Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III:

Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Why We Need Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.

Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Why We Need Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.

UNIT-IV:

Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric.

Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.

UNIT-V:

Blockchain Case Studies: Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities.

Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

TEXT BOOK:

1. Block Chain Technology, Chandramouli Subramanian, Asha A. George, Abhilash K. A. and Meena Karthikeyan, Universities Press

REFERENCES:

1. Blockchain Blueprint for Economy, Melanie Swan, SPD O'Reilly
2. Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gaur, Pearson Addison Wesley

ONLINE RESOURCES:

1. The Bitcoin Backbone Protocol - Analysis and Applications, J. A. Garay et al., EUROCRYPT2015LNCSVOI9057, Vol. II, pp. 281310 (eprint.iacr.org/2016/1048)

B.Tech. VIII Semester

(22PE1IT403) ADVERSARIAL MACHINE LEARNING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PREREQUISITES: Basic knowledge of Machine Learning

COURSE OBJECTIVES:

- To understand the basic notation and introduce the framework for secure learning and the various phases involved in the learning process
- To study causative attacks on machine learning algorithms and their impact on learning
- To examine a case study on causative attacks against the SpamBayes spam filter and understand the defense mechanisms used
- To explore different types of attacks on specific machine learning models such as hypersphere learners and PCA detectors
- To delve into exploratory attacks on machine learning, focusing on privacy-preserving mechanisms for SVM learning

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic notation and how security analysis fits into the learning process

CO-2: Apply the gained knowledge to defend machine learning models against causative attacks

CO-3: Create a practical experience in corrupting PCA subspaces

CO-4: Evaluate the SpamBayes spam filter and its susceptibility to causative attacks

CO-5: Analyse privacy-preserving mechanisms for SVM learning in protecting sensitive data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	1	1	2	1	2	-	-	-	1	2	1	2
CO-2	2	2	3	2	2	3	1	3	-	-	-	2	3	2	3
CO-3	2	2	2	3	3	1	2	2	-	-	-	2	3	3	2
CO-4	2	2	3	2	2	2	1	3	-	-	-	2	2	3	2
CO-5	2	2	3	2	2	2	1	1	-	-	-	2	3	2	3

UNIT-I:

Introduction

Background and Notation: Basic Notation, Statistical Machine Learning.

A Framework for Secure Learning: Analysing the Phases of Learning, Security Analysis, Framework, Exploratory Attacks, Causative Attacks.

UNIT-II:**Causative Attacks on Machine Learning**

Attacking a Hypersphere Learner: Causative Attacks on Hypersphere Detectors, Hypersphere Attack Description, Optimal Unconstrained Attacks, Imposing Time Constraints on the Attack, Constrained Attacks.

UNIT-III:

Availability Attack Case Study: Spam Bayes: The SpamBayes Spam Filter, Threat Model for SpamBayes, Causative Attacks against SpamBayes' Learner, the Reject on Negative Impact (RONI) Defense, Experiments with SpamBayes.

UNIT-IV:

Integrity Attack Case Study: PCA Detector: PCA Method for Detecting Traffic Anomalies, Corrupting the PCA Subspace, Corruption-Resilient Detectors, Empirical Evaluation.

UNIT-V:**Exploratory Attacks on Machine Learning**

Privacy-Preserving Mechanisms for SVM Learning: Privacy Breach Case Studies, Problem Setting: Privacy-Preserving Learning, Support Vector Machines: A Brief Primer.

TEXT BOOK:

1. Adversarial Machine Learning, Anthony D. Joseph, Blaine Nelson, Benjamin I. P. Rubinstein, J. D. Tygar, Cambridge University Press, 2019

REFERENCES:

1. Adversarial Machine Learning, Vol. 12, Vorobeychik, Yevgeniy, Murat Kantarcioglu, Ronald Brachman, Peter Stone, and Francesca Rossi, San Rafael, Morgan & Claypool Publishers, 2018
2. Generative Deep Learning: Teaching Machines to Paint, Foster David, Write, Compose, and Play (Japanese Version) O'Reilly Media, 2019,139-140

ONLINE RESOURCES:

1. <https://www.udacity.com/blog/2023/03/the-role-of-ai-and-ml-in-cybersecurity.html>
2. <https://www.youtube.com/watch?v=3-qazNQS2JU>

B.Tech. VIII Semester

(22PE1AM406) GENERATIVE AI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To familiarize with generative models and their role in AI
- To develop expertise in text generation techniques and frameworks
- To master image generation with advanced techniques
- To apply generative models to create Art, Music, and Interactive Systems
- To gain Proficiency with Open-Source Generative Tools and Deployment

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concepts of generative modelling

CO-2: Apply Gen AI to generating texts

CO-3: Understand Gen AI for generating video

CO-4: Apply Gen AI for generating video

CO-5: Apply open-source tools for solving problems using Gen AI

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	2	2	-	1	2	-	-	2	3	3	3
CO-2	3	3	3	1	2	2	-	1	2	-	-	2	3	3	3
CO-3	3	3	3	1	2	2	-	1	2	-	-	2	3	3	3
CO-4	3	3	3	1	2	2	-	1	2	-	-	2	3	3	3
CO-5	3	3	3	1	2	2	-	1	2	-	-	2	3	3	3

UNIT-I:

Introduction to GEN AI: Historical Overview of Generative modeling - Difference between Gen AI and Discriminative Modeling – Importance of generative models in AI and Machine Learning – Types of Generative models – GANs, VAEs, autoregressive models and Vector quantized Diffusion models - Understanding of probabilistic modeling and generative process - Challenges of Generative Modeling – Future of Gen AI – Ethical Aspects of AI – Responsible AI – Use Cases.

UNIT-II:

Generative Models for Text: Language Models Basics – Building blocks of Language models - Transformer Architecture – Encoder and Decoder – Attention mechanisms - Generation of Text – Models like BERT and GPT models – Generation of Text - Autoencoding – Regression Models – Exploring ChatGPT – Prompt Engineering – Designing Prompts– Revising Prompts using Reinforcement Learning from Human Feedback (RLHF) - Retrieval Augmented Generation – Multimodal LLM – Issues of LLM like hallucination.

UNIT-III:

Generation of Images: Introduction to Generative Adversarial Networks – Adversarial Training Process – Nash Equilibrium – Variational Autoencoders – Encoder-Decoder Architectures - Stable Diffusion Models – Introduction to Transformer-based Image Generation – CLIP – Visual Transformers ViT- Dall-E2 and Dall-E3, GPT-4V – Issues of Image Generation models like Mode Collapse and Stability.

UNIT-IV:

Generation of Painting Music and Play: Variants of GAN – Types of GAN - Cyclic GAN – Using Cyclic GAN to Generate Paintings – Neural Style Transfer – Style Transfer - Music Generating RNN – MuseGAN – Autonomous agents – Deep Q Algorithm – Actor-critic Network.

UNIT-V:

OpenSource Models and Programming Frame Works: Training and Fine tuning of Generative models – GPT4All - Transfer learning and Pretrained models - Training vision models – Google Copilot - Programming LLM – LangChain – Open Source Models – Llama - Programming for TimeSformer – Deployment – Hugging Face

TEXT BOOK:

1. Transformers for Natural Language Processing and Computer Vision, Denis Rothman, 3rd Edition, Packt Books, 2024

REFERENCES:

1. Generative Deep Learning, David Foster, O'Reilly, 2024
2. Generative AI for Everyone, Altaf Rehmani, BlueRose One, 2024

B.Tech. VIII Semester

(22PE1DS406) QUANTUM COMPUTING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basics of quantum computation and quantum mechanics
- To apply different models of quantum computation
- To illustrate the quantum models using machine learning algorithms
- To explore various Quantum algorithms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explore the quantum computation environment

CO-2: Identify the implications of quantum computation in the field of mathematics and associated fields

CO-3: Demonstrate building blocks and programming methodologies for quantum computing

CO-4: Apply quantum computation algorithms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	1	1	1	1	1	-	-	-	1	1	-	1
CO-2	2	3	3	2	1	1	1	1	1	-	1	1	2	-	1
CO-3	2	2	2	1	1	1	1	1	-	-	1	1	1	1	1
CO-4	2	2	1	1	1	1	1	1	-	-	-	1	1	-	1
CO-5	1	3	3	2	2	1	1	1	1	-	1	1	2	1	1

UNIT-I:

Introduction and Background: Overview, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation. A Preview of Quantum Physics.

Linear Algebra and The Dirac Notation: The Dirac Notation and Hilbert, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators

UNIT-II:

Qubits and The Framework of Quantum Mechanics: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems.

A Quantum Model of Computation: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations.

UNIT-III:

Superdense Coding and Quantum Teleportation: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.

Quantum Error Correction: Error Models for Quantum Computing, Encoding, Error Recovery. Architecture: Bits and Qubits, Classical Gates, Reversible Gates, Quantum Gates.

UNIT-IV:

Programming Languages: Programming in a Quantum World, Quantum Assembly Programming, Toward Higher-Level Quantum Programming.

Theoretical Computer Science: Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations.

UNIT V:

Quantum Algorithms: Deutsch's Algorithm, The Deutsch–Jozsa Algorithm, Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm

TEXT BOOKS:

1. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press
2. Quantum Computing for Computer Scientists, Noson S. Yanofsky, Michael A. Mannucci, Cambridge University Press

REFERENCES:

1. Quantum Computing, Ronald De Wolf, QuSoft, CWI and University of Amsterdam
2. Quantum Computing Explained, David McMahon, Wiley
3. Quantum Computing Fundamentals, William (Chuck) Easttom II

B.Tech. VIII Semester

(22PE1CY410) ETHICAL HACKING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Operating Systems, Computer Networks, Network Security & Cryptography

COURSE OBJECTIVES:

- To understand the basic concepts of ethical Hacking
- To learn the enumeration and system Hacking process
- To understand the threats of Trojans, backdoors, and sniffers
- To learn session hijacking and its various techniques, as well as DoS, DDoS, and botnet attacks
- To learn to defend against database attacks and understand social engineering techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand ethical hacking and footprinting process

CO-2: Analyze various enumeration and system hacking methods

CO-3: Identify, analyze, and counter cybersecurity risks posed by Trojans, backdoors, and sniffers

CO-4: Apply various techniques and methods used in session hijacking, Dos, DDOS and Botnets

CO-5: Detect and mitigate SQL injection vulnerabilities and implement effective policies to counter social engineering risks

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	-	3	-	3	-	-	3	3	3	-	2	-	-	3
CO-2	2	1	3	2	3	3	3	3	2	-	2	2	2	2	3
CO-3	2	2	3	3	3	3	2	3	1	-	2	2	2	2	3
CO-4	-	2	3	3	3	3	2	3	-	-	2	2	2	2	3
CO-5	-	-	3	2	2	3	3	3	3	3	2	2		3	3

UNIT-I:

Ethical Hacking Basics: Security Fundamentals, Security Testing, Hacker and Cracker Descriptions, Ethical Hackers, Test Plans—Keeping It Legal, Ethics and Legality, The Attacker's Process, The Ethical Hacker's Process,

Footprinting and Scanning: The Seven-Step Information-Gathering Process, Identifying Active Machines, Finding Open Ports and Access Points, OS Fingerprinting.

UNIT-II:

Enumeration: Windows Enumeration, Windows Security, NetBIOS and LDAP Enumeration, SNMP Enumeration, Linux/UNIX Enumeration, NTP Enumeration, SMTP Enumeration,

System Hacking: Nontechnical Password Attacks, Technical Password Attacks, Password Guessing, Automated Password Guessing. Password Sniffing, Keystroke Loggers, Privilege Escalation and Exploiting Vulnerabilities, Exploiting an Application, Buffer Overflow, Owning the Box, Cracking the Passwords, Hiding Files and Covering Tracks, File Hiding.

UNIT-III:

Trojans and Backdoors: Trojan Types, Trojan Ports and Communication Methods, Trojan Goals, Trojan Infection Mechanisms ,Effects of Trojans, Trojan Tools, Distributing Trojans, Trojan Tool Kits, Covert Communication and Tools, Port Redirection, Other Redirection and Covert Tools ,Keystroke Logging and Spyware ,Hardware, Software, Spyware, Trojan and Backdoor Countermeasures

Sniffers: Passive Sniffing, Active Sniffing, Address Resolution Protocol, ARP Poisoning and Flooding, Tools for Sniffing, Wireshark, Other Sniffing Tools, Sniffing and Spoofing Countermeasures

UNIT-IV:

Session Hijacking: Transport Layer Hijacking, Predict the Sequence Number, Take One of the Parties Offline, Take Control of the Session, Application Layer Hijacking, Session Sniffing, Predictable Session Token ID, Man-in-the-Middle Attacks, Man-in-the-Browser Attacks, Client-Side Attacks, Session-Hijacking Tools, Preventing Session Hijacking Denial of Service, Distributed Denial of Service, and Botnets: Types of DoS, Bandwidth Attacks, SYN Flood Attacks, Program and Application Attacks, Distributed Denial of Service, DDoS Tools, Botnets, Countermeasures of DoS, DDOS, and Botnet

UNIT-V:

Database Attacks Database Hacking, Identifying SQL Servers, SQL Injection Vulnerabilities, SQL Injection Hacking Tools.

Social Engineering: Types of Social Engineering, Person-to-Person Social Engineering, Computer-Based Social Engineering, Reverse Social Engineering, Policies and Procedures, Employee Hiring and Termination Policies, Help Desk Procedures and Password Change Policies, User Awareness.

TEXT BOOKS:

1. Certified Ethical Hacker (CEH) Cert guide, Michael Gregg, 3rd Edition, Pearson, 2019
2. Hands-On Ethical Hacking and Network Defense, Michael T. Simpson, Kent Backman, James E. Corley, 3rd Edition, Cengage Learning, 2017

REFERENCES:

1. Ethical Hacking and Countermeasures Attack Phases, EC-Council, Cengage Learning
2. CEH: Official Certified Ethical Hacker Review Guide, Kimberly Graves, Wiley, 2007

B.Tech. VIII Semester

(22PE1AM407) ROBOTIC PROCESS AUTOMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To explore the knowledge of basic concepts of Robotic Process Automation
- To outline the RPA design & development strategies
- To apply design methodologies adopted by UiPath
- To develop flow diagrams using RPA for real-world application
- To apply programming robots for logging and exception handling

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Robotic Process Automation concepts

CO-2: Adopt UiPath programming techniques to deploy robot configurations

CO-3: Explore various data extraction techniques and perform integrations with various popular applications

CO-4: Design and develop a programmed robot that includes logging and exception handling

CO-5: Deploy and control Bots with UiPath orchestrator

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	1	1	3	3	2	2	2	1	1	2	2	2	3	3
CO-2	2	2	2	1	3	1	1	2	1	1	1	1	2	2	2
CO-3	1	2	2	3	2	2	1	3	1	2	1	2	2	2	2
CO-4	2	2	3	3	2	2	1	2	1	2	1	2	2	1	2
CO-5	1	2	1	1	3	1	1	2	1	2	1	1	2	2	2

UNIT-I:

Introduction: What is Robotic Process Automation (RPA), Scope & techniques of Automation, Benefits of RPA, Components of RPA, RPA Platforms, UiPath Studio, Installation of UiPath Studio, Learning UiPath Studio.

UNIT-II:

Sequence, Flowchart & Control Flow: Sequencing the Workflow, Activities, Flowchart, Control Flow for Decision making. Data Manipulation: Variables, Collection, Arguments, Data Table, Clipboard management, File operations with step-by-step example

Controls: Finding the control, waiting for a control, Act on a control, UiExplorer, Handling Events Recording and advanced UI Interaction Basic recording, Desktop recording, Web recording, Citrix, Screen Scraping, when to use OCR, Types of OCR available, how to use OCR Avoiding typical failure points.

UNIT-III:

Plugins and Extensions: Terminal plugin, Java plugin, Java plugin with UiPath Studio, Citrix automation, Citrix environment, Mail plugin, PDF plugin, Web integration, Excel and Word plugins, Credential management Extensions Handling User Events and Assistant Bots: What are assistant bots, Monitoring system event triggers: Hotkey trigger, Mouse trigger, System trigger, Monitoring image and element triggers, Launching an assistant bot on a keyboard event.

UNIT-IV:

Exception Handling, Debugging, and Logging: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots Debugging techniques, setting break points, Slow step, Highlighting, Break, collecting crash dumps: Enabling crash dumps, Disabling crash dumps, Error reporting: Enterprise Edition customers, Community Edition users. Managing and Maintaining the Code: Project organization, picking an appropriate layout for each workflow, Breaking the process into smaller parts, Using exception handling, Making your workflow readable, Keeping it clean, Nesting workflows, Reusability of workflows, Templates, Commenting techniques, State Machine, When to use Flowcharts State Machines or Sequences.

UNIT-V:

Deploying and Maintaining the Bot: Publishing using publish utility, Overview of Orchestration Server, Using Orchestration Server to control bots, Using Orchestration Server to deploy bots, License management, Activating and uploading a license to Orchestrator, Publishing and managing updates, Packages, Managing packages.

TEXT BOOK:

1. Learning Robotic Process Automation, Alok Mani Tripathi, O'Reilly Publishing, 2018

REFERENCES:

1. Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant, Richard Murdoch, Amazon Asia-Pacific Holdings, 2018
2. Introduction to Robotic Process Automation: A Primer, Institute of Robotic Process Automation, Frank Casale, Rebecca Dilla, Heidi Jaynes, Lauren Livingston, Amazon Asia-Pacific Holdings, 2018

B.Tech. VIII Semester

(22PE1CS405) SOCIAL MEDIA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Graph Theory, Data Mining, Python/R programming

COURSE OBJECTIVES:

- To familiarize with the concept of social media
- To familiarize with the concept of social media analytics and understand its significance
- To enable to develop skills required for analyzing the effectiveness of social media
- To examine the ethical and legal implications of leveraging social media data
- To familiarize with different visualization techniques for Social media analytics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the concept of social media analytics and its significance

CO-2: Analyze the effectiveness of social media

CO-3: Use different effective visualization techniques to represent social media analytics

CO-4: Use different social media analytics tools effectively and efficiently

CO-5: Acquire the fundamental perspectives and hands-on skills needed to work with social media data

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	3	3	2	-	-	2	2	2	3	3	3	3
CO-2	3	2	3	2	3	3	-	-	2	2	2	3	3	3	3
CO-3	3	2	3	3	3	3	-	-	3	2	3	2	3	3	3
CO-4	3	2	2	3	2	2	-	-	3	2	2	3	3	2	2
CO-5	3	2	2	3	2	3	-	-	2	2	3	2	3	2	3

UNIT-I:

Social Media Analytics: An Overview: Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations. Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

UNIT-II:

Social Network Structure, Measures & Visualization: Basics of Social Network Structure - Nodes, Edges & Tie Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust Network Visualization - Graph Layout, Visualizing Network features, Scale Issues. Social Media Network Analytics - Common

Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

UNIT-III:

Social Media Text, Action & Hyperlink Analytics: Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text 8 Analysis Tools Social Media Action Analytics - What Is Actions Analytics? Common Social Media Actions, Actions Analytics Tools Social Media Hyperlink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

UNIT-IV:

Social Media Location & Search Engine Analytics: Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

UNIT-V:

Social Information Filtering: Social Information Filtering - Social Sharing and filtering, Automated Recommendation systems, Traditional Vs social Recommendation Systems Understanding SocialMedia and Business Alignment, Social Media KPI, Formulating a Social Media Strategy, Managing Social Media Risks

TEXT BOOKS:

1. Seven Layers of Social Media Analytics - Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data, Gohar F. Khan, 2015
2. Mining the Social Web - Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites, Matthew A. Russell, O'Reilly
3. Social Network Data Analytics, Charu Aggarwal (Ed.), Springer, 2011

REFERENCES:

1. Analyzing the Social Web, Jennifer Golbeck, 1st Edition
2. Social Media Analytics, Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press, 2015
3. Social Media Analytics Strategy - Using Data to Optimize Business Performance, Alex Gonçalves, APress
4. Social Media Data Mining and Analytics, Szabo G., G. Polatkan, O. Boykin & A. Chalkiopoulos, Wiley, 2019
5. Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, Marshall Sponder, 1st Edition, McGraw-Hill, 2011

ONLINE RESOURCES:

1. <https://cse.iitkgp.ac.in/~pawang/courses/SC16.html>
2. https://onlinecourses.nptel.ac.in/noc20_cs78/preview